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Summary

I am a postdoctoral researcher at the Geometry group of KU Leuven. I am primarily interested in the different ways in which special spinors encode geometric properties of manifolds. I am also passionate about teaching.

Experience

KU Leuven

Postdoctoral researcher
Research area: Differential Geometry

Leuven, Belgium
November 2025 – present

Education

Imperial College London

PhD in Mathematics
Research area: Differential Geometry
Supervisors: Marie-Amélie Lawn and Steven Sivek
Examiners: Simon Donaldson and Simon Salamon

London, UK
October 2021 – October 2025

University of Cambridge

MASt in Mathematics (Part III)

Cambridge, UK
September 2020 – June 2021

Autonomous University of Barcelona

Bachelor's Degree in Mathematics

Barcelona, Spain
September 2015 – June 2020

Autonomous University of Barcelona

Bachelor's Degree in Physics

Barcelona, Spain
September 2015 – June 2020

Scholarships

Roth Scholarship –from the Department of Mathematics at Imperial College London	2021
EPSRC Scholarship EP/W523872 –funding my doctoral studies	2021

Publications

On the stability of Einstein metrics carrying a special twisted spinor

Preprint, 2025 [Available here](#).

Killing Mean Curvature Solitons from Riemannian Submersions

with Marie-Amélie Lawn and Miguel Ortega.

Journal of Mathematical Analysis and Applications, 2026. [Available here](#).

Special spinors in generalised spin geometry

PhD thesis, 2025 [Available here](#).

Invariant Spinors on Flag Manifolds

with Uwe Semmelmann.

International Journal of Mathematics, 2025. [Available here](#).

The Geometry of Generalised Spin^r Spinors on Projective Spaces

with Jordan Hofmann.

Symmetry, Integrability and Geometry: Methods and Applications (SIGMA), 2025. [Available here](#).

Generalised Killing Spinors on Three-Dimensional Lie Groups

Manuscripta Mathematica, 2025 [Available here](#).

Generalised Spin^r Structures on Homogeneous Spaces

with Marie-Amélie Lawn.

Differential Geometry and its Applications, 2025. [Available here](#).

New Examples of Translating Solitons in Generalised Robertson-Walker Geometries

with Marie-Amélie Lawn and Miguel Ortega.

Asian Journal of Mathematics, 2025. [Available here](#).

Research stays

Research stay at the University of Granada

September 2024

Collaboration with Miguel Ortega, developing new methods for the study of mean curvature flow solitons using Riemannian submersions.

Research stay at Phillips Universität Marburg

November 2023

Collaboration with Ilka Agricola, initiating a project on special contact metric structures.

Research stay at the University of Granada

September 2022

Collaboration with Miguel Ortega, studying mean curvature flow on generalised Robertson-Walker spacetimes.

Undergraduate research stay at the Autonomous University of Barcelona

2017

I studied some aspects of non-commutative algebra under the supervision of Francesc Perera.

Research stay at the Institute for High Energy Physics of Barcelona (IFAE)

July 2016

Teaching

Associate Fellowship of the Higher Education Academy

June 2024 – Present

Official recognition of my teaching abilities and experience.

Supervision of undergraduate research project

December 2024 – May 2025

Supervision of a second-year undergraduate from the University of Bristol, working on Killing spinors with respect to the canonical connection on a homogeneous space.

Senior Graduate Teaching Assistant at Imperial College

October 2022 – October 2025

Courses: Introduction to University Mathematics, Linear Algebra and Groups.

Among my duties are marking, checking academic materials, coordinating the teaching task of my fellow teaching assistants, monitoring problem sessions and demonstrating at the whiteboard.

Graduate Teaching Assistant at Imperial College

October 2021 – October 2025

Courses: Introduction to University Mathematics, Linear Algebra and Groups, Analysis I, Analysis II, Measure Theory, Lebesgue Measure and Integration, Groups and Rings.

Undergraduate Teaching Assistant at the Autonomous University of Barcelona

2018

Courses: Linear Algebra.

I monitored some problem-solving sessions for first-year undergraduate students.

Talks

Generalised Spin Structures

September 2025

Differential Geometry and its Applications. Brno, Czech Republic.

[Link to event site](#).

Generalised Spin Structures

July 2025

Workshop on Even and Odd Dimensional Geometric Structures. Bari, Italy.

[Link to event site](#).

A Generalisation of Spin Structures

May 2025

Seminari de Topologia de la UB. Barcelona, Spain.

[Link to event site](#).

Quaternion-Kähler Manifolds

February 2025

Quaternion-Kähler manifolds and the LeBrun-Salamon Conjecture. The third Marburger Arbeitsgemeinschaft Mathematik (MAM III). Marburg, Germany.

[Link to event site.](#)

Spin Geometry and Generalisations

November 2024

Seminari de Geometria de la UAB. Barcelona, Spain.

[Link to event site.](#)

Characteristic Classes on Homogeneous Spaces

June 2024

Spin Geometry Working Group. London, UK.

What is Spin Geometry?

January 2024

KCL/UCL Junior Geometry Seminar. London, UK.

[Link to seminar site.](#)

Twisted Spin Structures

November 2023

Forschungsseminar Differentialgeometrie und Analysis. Marburg, Germany.

[Link to seminar site.](#)

Invariant Twisted Spinors on Homogeneous Spaces

October 2023

The Crazy World of Arthur L. Besse. A workshop on Einstein manifolds. Stuttgart, Germany.

[Link to event site.](#)

Generalised Spin^r Structures on Homogeneous Spaces

May 2023

BIRS Workshop: Spinorial and Octonionic Aspects of G_2 and Spin(7) Geometry. Banff, Canada.

[Link to event site.](#) [Recorded talk.](#)

Tauberian Operators on Banach Spaces

March 2020

Red de Análisis Funcional y Aplicaciones: X Workshop on Functional Analysis in Tenerife, Spain.

[Link to event site.](#)

Referees

Prof. Dr. Uwe Semmelmann

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Prof. Miguel Ortega

miortega@ugr.es

Dr. Marie-Amélie Lawn

m.lawn@imperial.ac.uk

Dr. Charlotte Kestner (teaching)

c.kestner@imperial.ac.uk